

RAPHAEL SULZER



CONTACT

✉ raphael.sulzer.1@gmail.com
🏠 raphaelsulzer.de
📍 Nice, France

🌐 LinkedIn
🐙 GitHub
📄 Google Scholar
🔍 ResearchGate

SKILLS

Programming

Python	9 yrs
C++	8 yrs
Bash	5 yrs
HTML/CSS	4 yrs
C#	2 yrs
JavaScript	2 yrs

Languages

German	Native
English	Fluent
French	Conversational
Dutch	Beginner

Other

3D modeling aws Blender

CGAL CMake computer vision

conda deep learning Docker

GIS geometry processing

GitHub LaTeX LiDAR

machine learning matplotlib

mesh nanobind numpy

photogrammetry point cloud

pytorch unix vscode

EXPERIENCE

RESEARCH ENGINEER

📅 11/2023 - Present
📍 LUXCARTA, MOUANS-SARTOUX, FRANCE

Research and development of 2D and 3D mapping solutions using machine learning and geometry processing techniques applied to remote sensing data.

POSTDOCTORAL RESEARCHER

📅 11/2022 - Present
📍 TITANE, INRIA, SOPHIA-ANTIPOLIS, FRANCE

Research at the intersection of geometry processing, computer vision, and deep learning.

PHD RESEARCHER

📅 12/2018 - 10/2022
📍 LASTIG, INSTITUT GÉOGRAPHIQUE NATIONAL, PARIS, FRANCE
📍 IMAGINE, ÉCOLE DES PONTS PARISTECH, MARNE-LA-VALLÉE, FRANCE

Research at the intersection of geometry processing, computer vision, and deep learning.

LECTURER

📅 12/2018 - 12/2019
📍 ÉCOLE NATIONALE DES SCIENCES GÉOGRAPHIQUES, PARIS, FRANCE

Design and implementation of e-learning courses in photogrammetry and GIS.

GIS DEVELOPER

📅 05/2018 - 07/2019
📍 GEO-COL GIS AND COLLABORATIVE PLANNING, AMSTERDAM, NETHERLANDS

Development of custom GIS applications for public and private customers.

GRADUATE STUDENT INTERN

📅 06/2017 - 02/2018
📍 ARUP, AMSTERDAM, NETHERLANDS

Development of an algorithm for building classification from remote sensing and cadastral data.

EDUCATION

PHD DEGREE, DEEP LEARNING AND GEOMETRY PROCESSING

📅 12/2018 - 10/2022
📍 GUSTAVE EIFFEL UNIVERSITY, MARNE-LA-VALLÉE, FRANCE
PhD Thesis: Learning Surface Reconstruction from Point Clouds in the Wild ([link](#))

MASTER'S DEGREE, MSC GEOMATICS, CUM LAUDE

📅 08/2016 - 05/2018
📍 DELFT UNIVERSITY OF TECHNOLOGY, DELFT, NETHERLANDS
Master's Thesis: Shape Based Classification of Seismic Building Structural Types ([link](#))

ERASMUS EXCHANGE SEMESTER

📅 08/2015 - 06/2016
📍 DELFT UNIVERSITY OF TECHNOLOGY, DELFT, NETHERLANDS

MSC GEODESY AND GEOINFORMATICS

📅 05/2015 - 01/2016
📍 UNIVERSITY OF STUTTGART, STUTTGART, GERMANY

BACHELOR'S DEGREE, BSC GEODESY AND GEOINFORMATICS

📅 10/2011 - 05/2015
📍 UNIVERSITY OF STUTTGART, STUTTGART, GERMANY
Bachelor's Thesis: Photogrammetric Measurement of Snow Depth Using an UAV Platform ([link](#))

PUBLICATIONS

The P³ Dataset: Pixels, Points and Polygons for Multimodal Building Vectorization

 **R. Sulzer**, L. Duan, N. Girard, F. Lafarge

 2025  preprint

 [arXiv](#)

Concise Plane Arrangements for Low-Poly Surface and Volume Modelling

 **R. Sulzer**, F. Lafarge

 2024  European Conference on Computer Vision (ECCV)

 [arXiv](#)

SimpliCity: Reconstructing Buildings with Simple Regularized 3D Models

 J.-P. Bauchet, **R. Sulzer**, F. Lafarge, Y. Tarabalka

 2024  Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

 [arXiv](#)

A Survey and Benchmark of Automatic Surface Reconstruction from Point Clouds

 **R. Sulzer**, L. Landrieu, R. Marlet, B. Vallet

 2024  IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

 [arXiv](#)

Evaluating Surface Mesh Reconstruction Using Real Data

 Y. Marchand, L. Caraffa, **R. Sulzer**, E. Clédat, B. Vallet

 2023  Photogrammetric Engineering & Remote Sensing (PE&RS)

 [link](#)

Learning Surface Reconstruction from Point Clouds in the Wild

 **R. Sulzer**

 2022  Université Gustave Eiffel

 [link](#)

Deep Surface Reconstruction from Point Clouds with Visibility Information

 **R. Sulzer**, L. Landrieu, A. Boulch, R. Marlet, B. Vallet

 2022  26th International Conference on Pattern Recognition (ICPR)

 [arXiv](#)

Scalable Surface Reconstruction with Delaunay-Graph Neural Networks

 **R. Sulzer**, L. Landrieu, R. Marlet, B. Vallet

 2021  Computer Graphics Forum (CGF)

 [arXiv](#)

Shape Based Classification of Seismic Building Structural Types

 **R. Sulzer**, P. Nourian, M. Palmieri, J. van Gemert

 2018  ISPRS International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences

 [link](#)

Track-id: Activity Determination based on Wi-Fi Monitoring

 van der Spek, S., Verbree, E., Quak, W., Groeneveld, I. J. D. G., **Sulzer, R.**, Theocharous, E., Xu, Y.

 2016  13th International Conference on Location-Based Services (LBS)

 [link](#)

SERVICES

Journal reviewing

- IEEE Transactions on Graphics
- IEEE Transactions on Visualization and Computer Graphics
- ISPRS Journal of Photogrammetry and Remote Sensing
- Computer Graphics Forum

Conference reviewing

- ACM SIGGRAPH
- IEEE Conference on Computer Vision and Pattern Recognition

REFERENCES

Florent LAFARGE

 florent.lafarge@inria.fr

 TITANE, INRIA

 Postdoctoral advisor

Loïc LANDRIEU

 loic.landrieu@enpc.fr

 IMAGINE, ÉCOLE DES PONTS PARISTECH

 Doctoral advisor